

Unit-II
Lecture: 1
Searching

A fundamental operation of a computer is to store huge information and retrieve information as quick as possible. Searching plays a pivotal role in the task of information retrieval.

- Searching methods are governed with **how data are stored in computer either in internal memory or in external memory.**
- Searching methods are also **decided by the data structures used to store information.**
- Searching methods are also **decided by search over a small set of data to a large set of data.**

Searching in general is the most time consuming task in many applications and the application of a good searching method eventually leads to a substantial increase in performance.

Searching methods can be broadly classified into two categories:

- **Internal Search**
 - A searching method is termed as internal search that is searching involved with data, which are **entirely stored in the internal memory of a computer.**
 - All internal searching methods are of two types:
 - with key comparisons and
 - without any comparisons.
 - There are two sub class of searching with key comparisons:
 - **Linear search** when data are **stored with linear data structures like array**, linked list etc.
 - **Non linear search** where data are stored with **non-linear data structures like tree, graphs etc.**
- **External Search**
 - External searching methods deal **with a large volume of data and stored in external storage device like magnetic tapes, disks etc.**

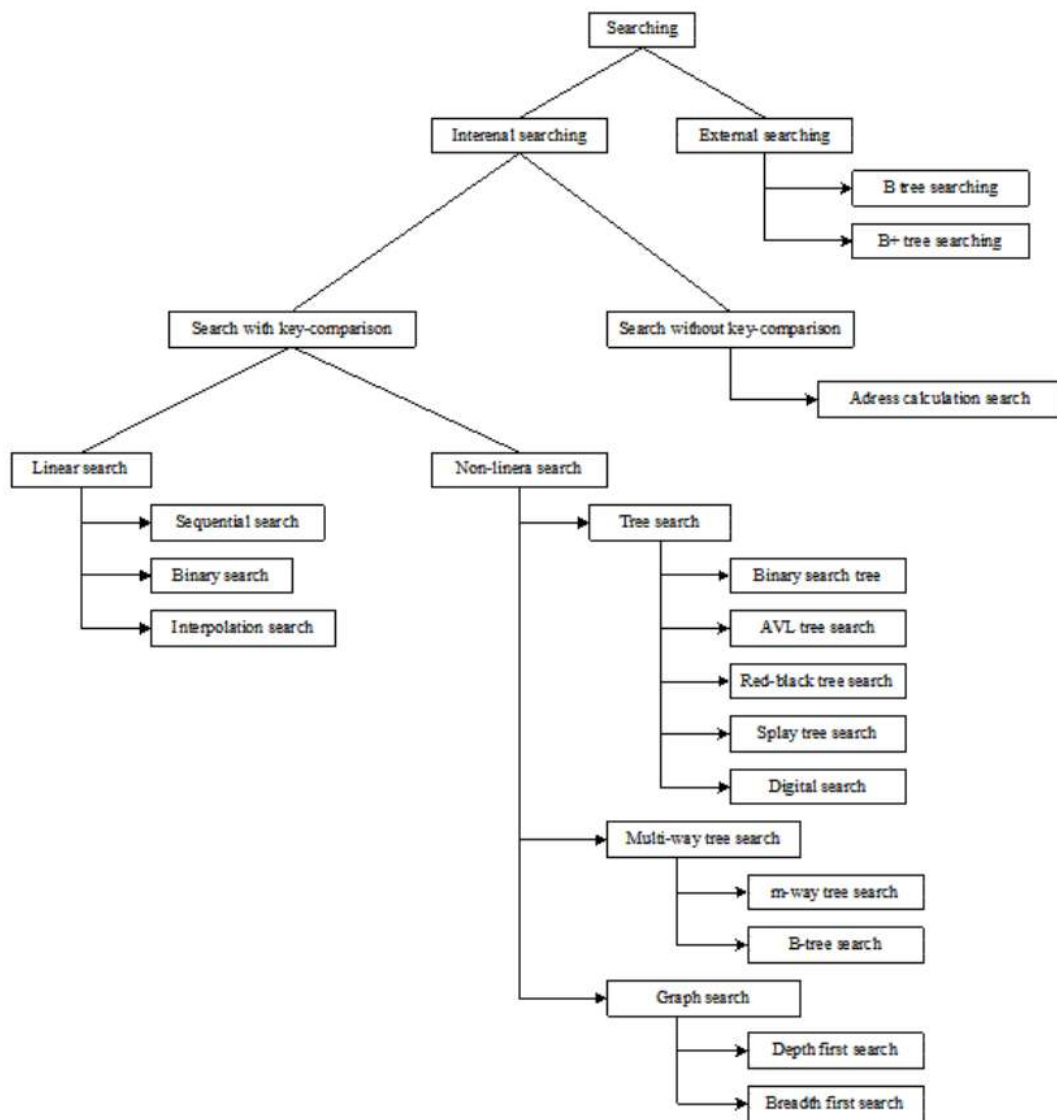


Fig: Classification of searching methods

Linear Search:

The simplest search technique is the sequential search/ linear search.

- In this technique, we start at the beginning of the list/array and search for the desired element by matching the value with each record being traversed, the process is continued till either the desired record is found or list is completely traversed.

e.g. Consider the following array A:

23 15 18 17 42 96 103

Now, let us search for 17 by linear search.

- The searching starts from the first position Since $A[0] \neq 17$,
- the search proceeds to the next position i.e. second position $A[1] \neq 17$
- The above process continues until the search element is found such as $A[3] = 17$
- Hence the searching element is found in the position 4.

Advantages:

- It is the simplest known technique.
- The elements in the list can be in any order.

Disadvantages:

- This method is inefficient when large number of elements is present in list because time taken for searching is more.

The worst case time complexity of linear search is $O(n)$.